

RETAIL SALES & PROFIT INTELLIGENCE DASHBOARD

E-Commerce Analytics Project | Data Cleaning, Analysis & Visualization with Excel
Module-End Assignment — DA Module 1

Student Name	Batch	Course
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Total Records	Analysis Period	Tables Used
2,000 Sales Transactions	Oct 2023 — Oct 2025	4 (Sales, Customer, Product, Store)

1. PROJECT OVERVIEW

Project Objective

This capstone project demonstrates end-to-end data analytics on an e-commerce retail dataset using Microsoft Excel. The workflow covers data cleaning and transformation, table integration using XLOOKUP/VLOOKUP, Excel formulas for business calculations, descriptive statistical analysis, Pivot Table summarization, and an interactive dashboard with charts for data visualization and insight generation.

- Clean and organize raw data across 4 dimension tables
- Handle missing and inconsistent values with appropriate strategies
- Integrate multiple tables using XLOOKUP/VLOOKUP functions
- Calculate key metrics: Sales, Cost, Profit, and Discount Value
- Perform descriptive statistical analysis using Analysis ToolPak
- Create Pivot Tables for multi-dimensional business summarization
- Develop an interactive dashboard with charts and slicers
- Generate actionable business insights and strategic recommendations

Dataset Overview

Table Name	Records	Columns	Description
Sales_Fact	2,000	10	Core transaction data
Customer_Dim	500	8	Customer demographics & loyalty
Product_Dim	100	7	Products, categories & costs
Store_Dim	20	5	Store regions & types

2. DATA CLEANING & TRANSFORMATION

2.1 Missing Value Treatment

Table	Column	Missing	Method	Formula / Logic
Customer_Dim	Loyalty_Level	4	Fill: UNKNOWN	=IF(ISBLANK(H2),"UNKNOWN",H2)
Product_Dim	Cost	6	Fill: Average	Average replacement (\$254.04)
Product_Dim	Stock	3	Fill: Median	Median imputation
Sales_Fact	Quantity	5	Fill: Median	Median imputation
Sales_Fact	Unit_Price	3	Fill: Mean	Mean imputation
Sales_Fact	Discount	4	Fill: 0	No discount assumed
Sales_Fact	Total_Amount	4	Recalculate	=Quantity*Unit_Price*(1-Discount)

2.2 Transformation & Calculated Fields

- Name Cleaning: Applied TRIM() and PROPER() to fix extra spaces and inconsistent casing in Customer names
- Age Group Segmentation: =IF(Age>=60,"Senior Citizen",IF(Age>=30,"Adult","Young")) — creates 3 customer age tiers
- Discount Value: =Unit_Price*Discount — monetary value of discount per transaction
- Total Cost: =Quantity*Cost — total cost per line item
- Total Profit: =Total_Amount-Total_Cost — net profit per transaction
- Table Integration: Used XLOOKUP to join Customer_Dim, Product_Dim, and Store_Dim onto Sales_Fact
- Currency & Percentage Formatting applied to all relevant columns
- Date Formatting: Standardized Order_Date to YYYY-MM-DD format

3. DESCRIPTIVE STATISTICS

Descriptive statistics were computed using the Excel Analysis ToolPak across all numerical fields including Unit_Price, Quantity, Discount, Total_Amount, Total_Cost and Total_Profit.

Statistic	Total Amount	Total Cost	Total Profit
Mean	\$1,085.45	\$640.04	\$445.41
Median	\$836.86	\$464.88	\$349.38
Standard Deviation	\$985.55	\$617.25	\$788.53
Minimum	\$26.21	-	-\$1,705.51
Maximum	\$5,992.04	-	\$3,625.65
Sum	\$2,170,907.46	\$1,280,568.61	\$890,338.85
Skewness	1.42	1.21	0.59

Kurtosis	1.95	1.48	0.71
Count	2,000	2,000	2,000

4. PIVOT TABLE ANALYSIS

4.1 Category-wise Sales & Profit

Category	Total Sales	Total Profit	Orders
Sports	\$542,010.17	\$287,778.41	498
Electronics	\$504,311.35	\$127,559.58	454
Home	\$427,295.80	\$191,039.75	410
Clothing	\$354,904.46	\$143,928.64	317
Beauty	\$299,812.74	\$136,384.48	282
TOTAL	\$2,128,334.52	\$886,691.26	1,961

4.2 Region-wise Profit Performance

Region	Total Sales	Total Profit	Orders
South ☐	\$591,741.62	\$273,292.68	537
North	\$498,397.28	\$207,684.47	459
East	\$558,618.17	\$239,534.45	510
West ☐	\$479,577.55	\$169,826.76	455

5. KEY BUSINESS INSIGHTS

TOTAL SALES \$2,170,907	TOTAL PROFIT \$890,339	TOTAL ORDERS 2,000	CUSTOMERS 500	PROFIT MARGIN 41.0%
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5.1 Summary Findings

- Total Sales Revenue of \$2,170,907 and Total Profit of \$890,339 represent a strong 41% profit margin across 2,000 orders
- South Region generated the highest profit at \$273,292 (30.7% of total profit) — recommended for further investment and expansion
- West Region generated the lowest profit at \$169,827 — marketing and operational improvements are needed here
- Sports Category leads in both Sales (\$542,010) and Profit (\$287,778) — maintain high inventory and marketing focus
- Electronics has high sales volume but lower profit margin — indicating higher costs or heavier discounting
- Customer Loyalty analysis shows Platinum and Gold tiers drive disproportionate sales value
- Monthly trend reveals demand peaks in March, June, September and November — align inventory accordingly
- Average Order Value is \$1,085 with average profit of \$445 per transaction

5.2 Business Questions Answered

#	Business Question	Answer
1	Which region generated the highest profit?	South Region — \$273,292.68
2	Which region generated the lowest profit?	West Region — \$169,826.76
3	Which category generated the highest sales?	Sports — \$542,010.17
4	What is the total sales revenue?	\$2,170,907.46
5	What is the total profit?	\$890,339 (41.0% margin)
6	How many orders were processed?	2,000 transactions
7	How many customers were analyzed?	500 unique customers
8	Which store type performs best?	Flagship stores lead in profitability
9	What is the average discount rate?	~14.8% average discount
10	Why was profit calculated?	To measure actual earnings after deducting product costs from revenue

6. STRATEGIC RECOMMENDATIONS

#	Recommendation	Business Rationale	Priority
1	Increase investment in South region	Highest profit region — scalable growth	❑ HIGH
2	Boost Sports category inventory	Top revenue and profit category	❑ HIGH
3	Revamp West region marketing strategy	Lowest profitability region	❑ MEDIUM
4	Optimize discount strategy to protect margins	High discount rate (14.8%) reduces profit	❑ HIGH
5	Strengthen customer loyalty program	Top tiers drive disproportionate revenue	❑ MEDIUM
6	Expand flagship store footprint	Flagship stores outperform other types	❑ MEDIUM
7	Align inventory with monthly demand peaks	Sales spikes in Mar/Jun/Sep/Nov	❑ TRACK

7. EXCEL FORMULAS & FUNCTIONS USED

Calculation	Formula	Purpose
Discount Value	=Unit_Price*Discount	Monetary value of discount per transaction
Total Cost	=Quantity*Cost	Total cost per sales line item
Total Profit	=Total_Amount-Total_Cost	Net profit per transaction after cost
Age Group	=IF(Age>=60,"Senior Citizen",IF(Age>=30,"Adult","Young"))	Segment customers into 3 age tiers
Missing Loyalty	=IF(ISBLANK(H2),"UNKNOWN",H2)	Handle null values in Loyalty_Level
XLOOKUP	=XLOOKUP(Cust_ID,Cust!A:A,Cust!B:B)	Join Customer dimension to Sales_Fact
Total Sales	=SUM(Total_Amount)	KPI: Overall revenue
Average Order Val	=AVERAGE(Total_Amount)	Mean transaction value KPI
Total Orders	=COUNT(Sales_ID)	KPI: Transaction volume
Profit Margin	=SUM(Total_Profit)/SUM(Total_Amount)	Overall profitability ratio
COUNTIF	=COUNTIF(Category,"Sports")	Count orders by category
IFERROR	=IFERROR(value,"N/A")	Handle formula error cases

9. CONCLUSION

This project successfully demonstrated a complete data analytics pipeline using Microsoft Excel on an e-commerce retail dataset. Starting from four raw tables with missing values and inconsistencies, the data was systematically cleaned, transformed, integrated, and analyzed to produce a comprehensive business intelligence dashboard.

The analysis revealed that Sports and Electronics are the highest-revenue categories, the South region leads in profitability, and Flagship stores outperform other store formats. These insights are supported by descriptive statistics, six pivot tables, and four interactive chart types embedded in the dashboard.

The project equips business stakeholders with data-driven intelligence to optimize inventory, refine pricing strategies, focus regional investment, and strengthen customer loyalty programs — all critical levers for sustainable revenue growth.

7. DATA DASHBOARD SUMMARY

This section presents a comprehensive visual summary of the Retail Sales Intelligence Dashboard, highlighting key performance indicators, category performance, regional profit breakdown, store-type analysis, and monthly sales trends. All figures are derived from the Excel Dashboard sheet.

KEY PERFORMANCE INDICATORS (KPIs)

□ TOTAL SALES \$2,174,395	□ TOTAL PROFIT \$886,691	□ TOTAL ORDERS 2,000	□ CUSTOMERS 500	□ AVG ORDER VALUE \$1,087	✓ PROFIT MARGIN 40.8%
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CATEGORY PERFORMANCE ANALYSIS

Category	Total Sales	Total Profit	Orders
□ Sports	\$542,010	\$287,778	498
□ Electronics	\$504,311	\$127,560	454
□ Home	\$427,296	\$191,040	410
□ Clothing	\$354,904	\$143,929	317
□ Beauty	\$299,813	\$136,384	282

REGIONAL PROFIT ANALYSIS

Region	Total Sales	Total Profit	Orders
□ South	\$635,213	\$269,332 ↑	581
North	\$646,101	\$254,966	596
East	\$539,031	\$253,305	510
⚠ West	\$354,050	\$109,088 ↓	313

STORE TYPE PERFORMANCE

Store Type	Total Sales	Total Profit
□ Flagship	\$848,577	\$351,587
□ Online □	\$914,254	\$363,759
□ Outlet	\$411,565	\$171,345

DASHBOARD INSIGHTS & CONCLUSIONS

Top Performers: Online Store leads all channels with \$914,254 in sales and \$363,759 profit. Sports category dominates with \$287,778 profit. South Region achieves highest regional profit at \$269,332.

Areas for Improvement: West Region lags behind at \$109,088 profit — targeted promotions and inventory adjustments recommended. Electronics shows high volume but lower profit margin vs. Sports category.

Overall Business Health: With a 40.8% profit margin across 2,000 orders from 500 customers, the retail business demonstrates strong financial health. Average order value of \$1,087 reflects high-value transactions. Monthly trend shows consistent demand peaks in March, June, September, and November — ideal periods for inventory build-up.